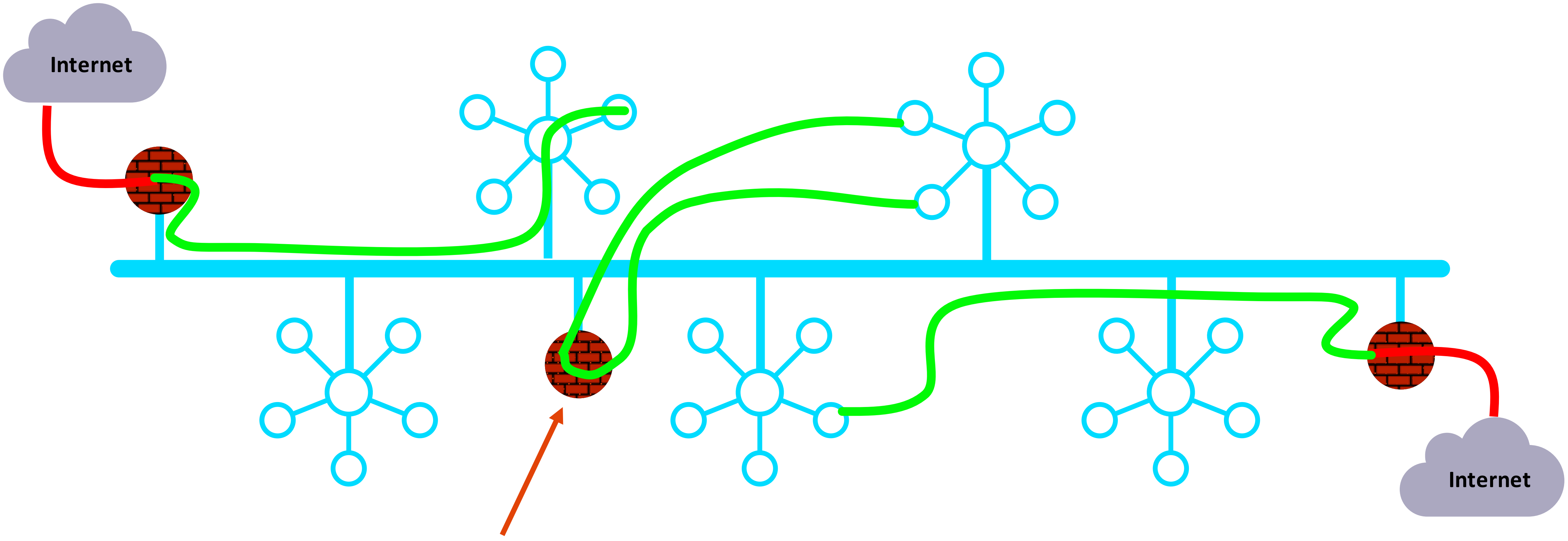
Distributed Cloud Firewall

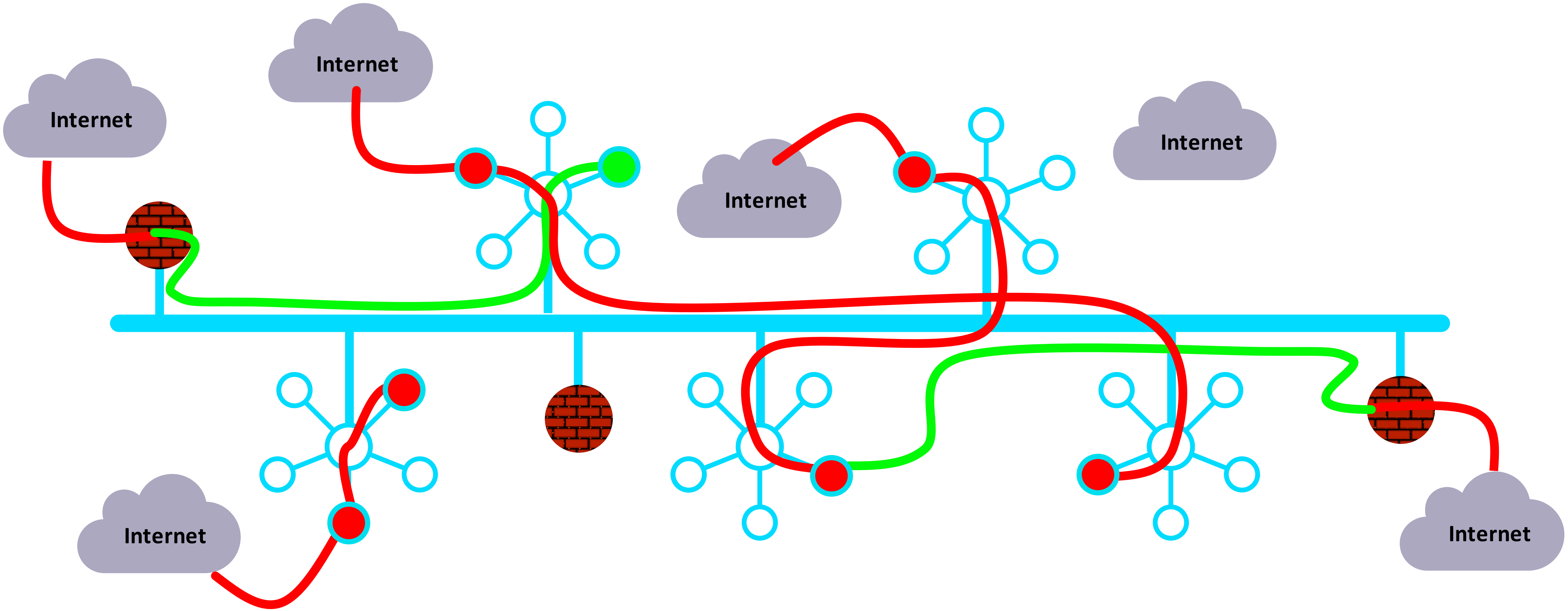
ACE Team

As Architected with Lift-and-Shift, Bolt-on, Data Center Era Products…



“Last Generation Firewalls”

In Reality…



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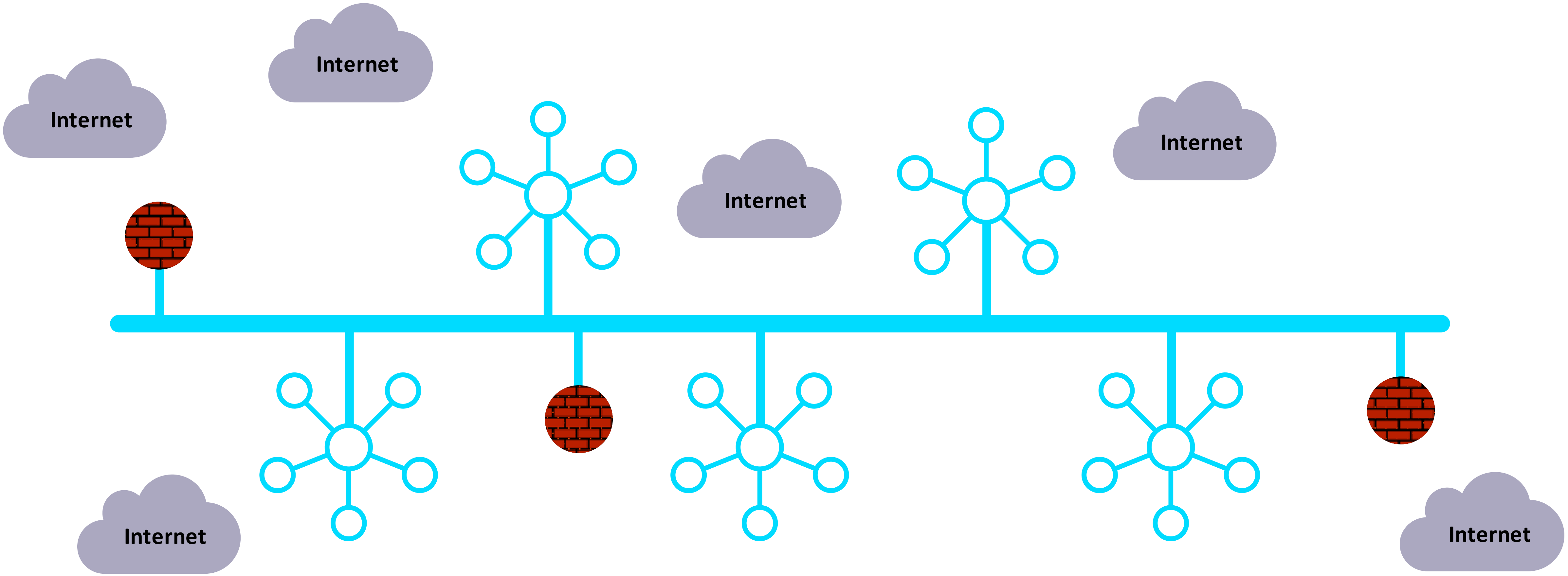
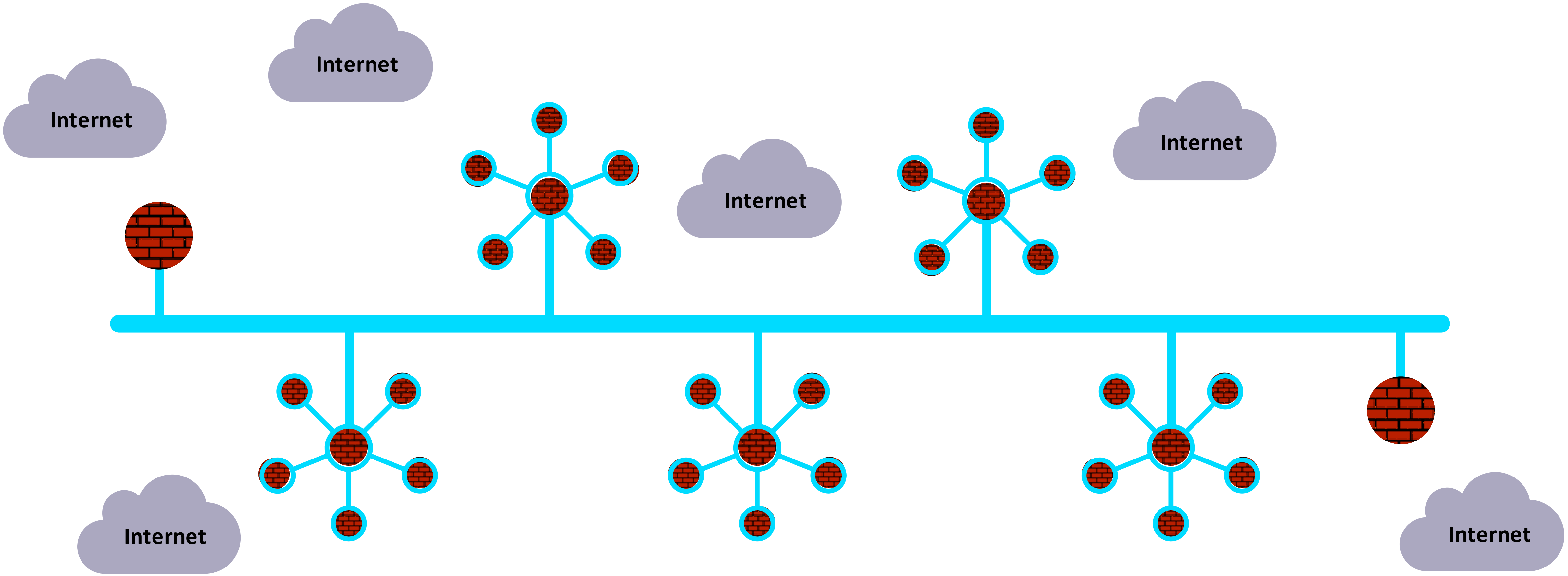
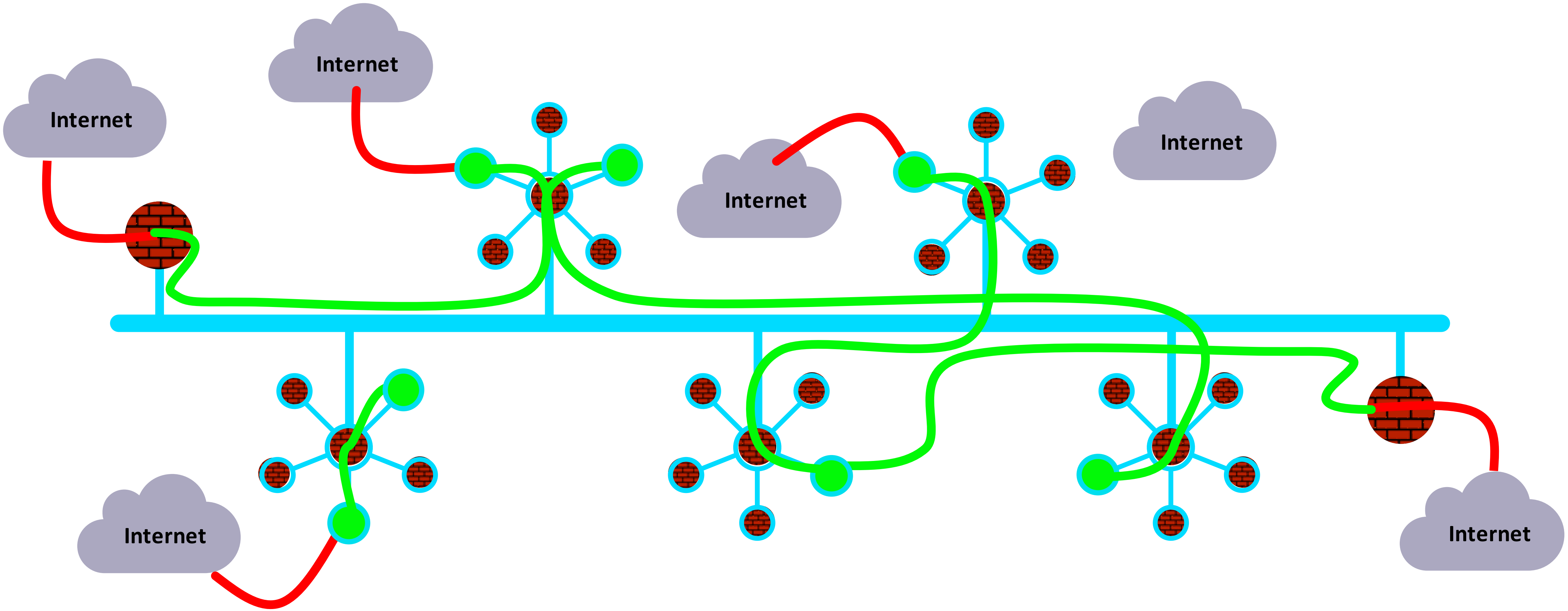
What If… the architecture was built for cloud



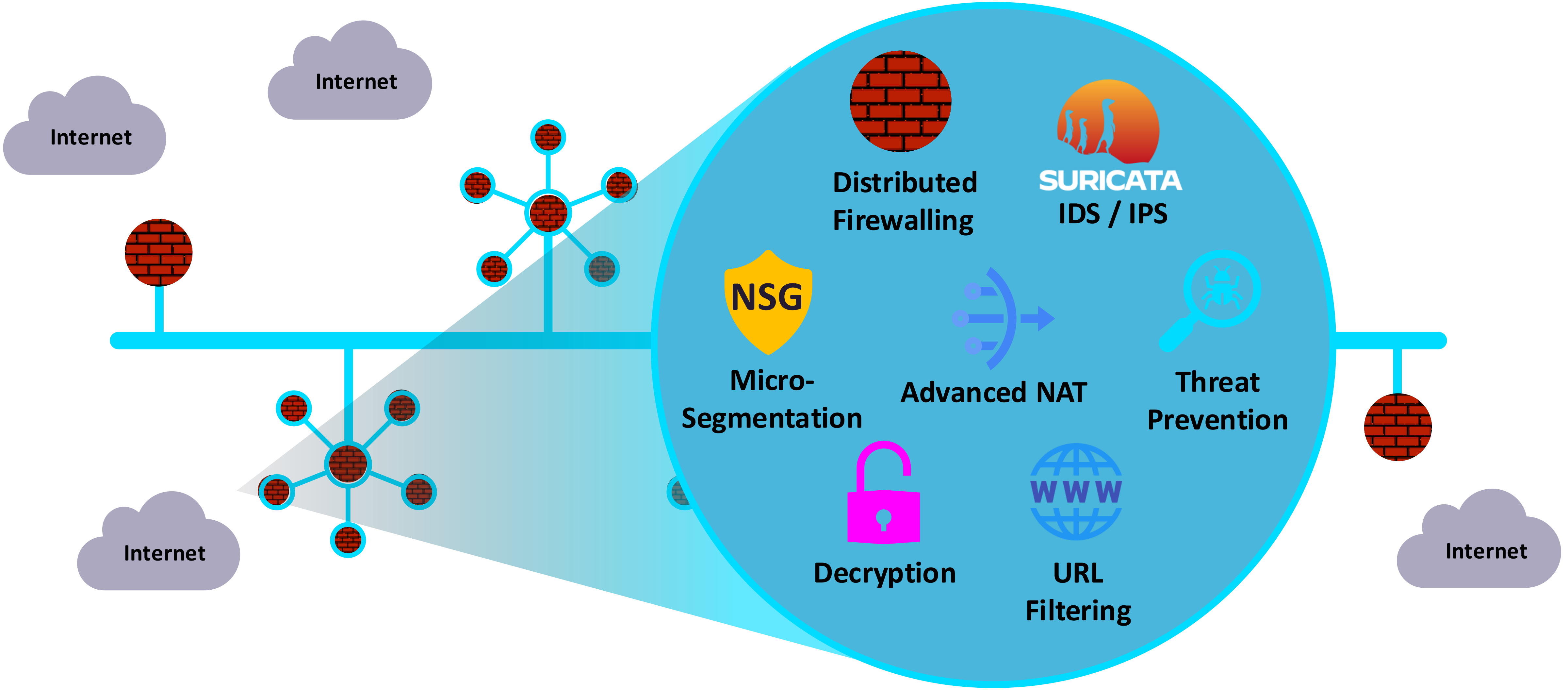
Firewalling Functions were Embedded in the Cloud Network Everywhere…



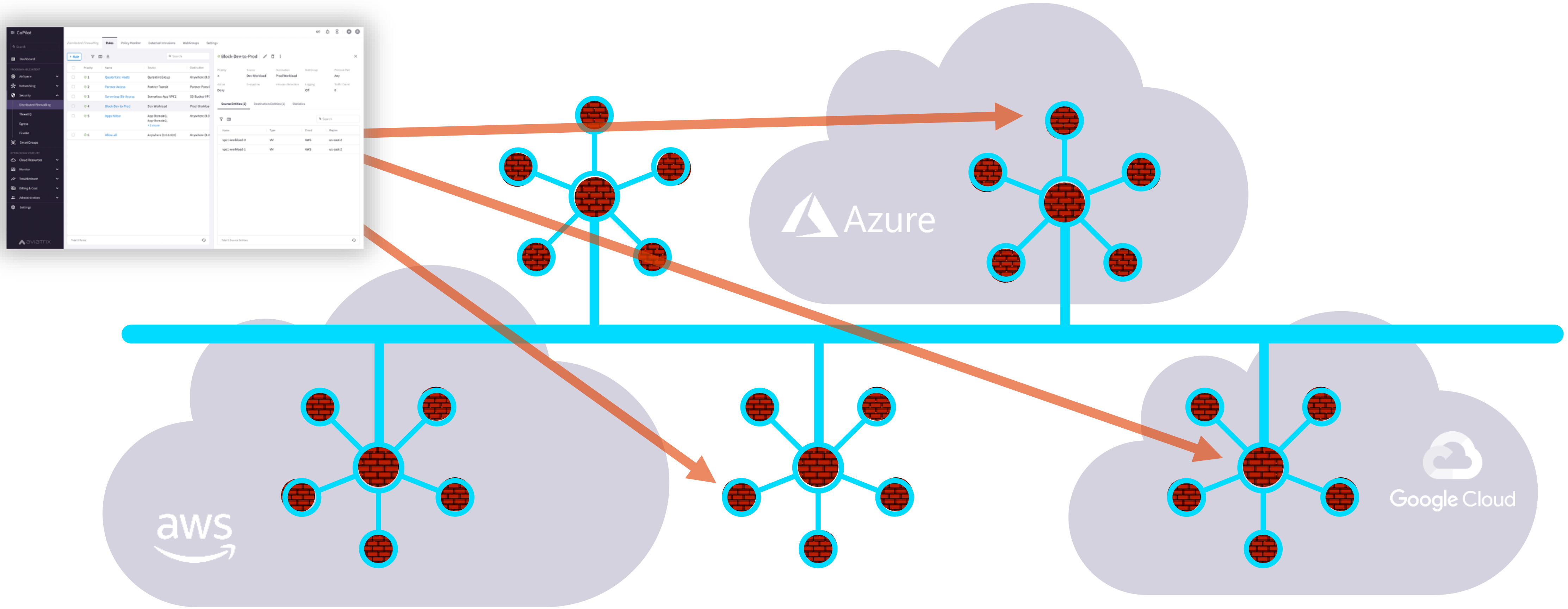
Centrally Managed, with Distributed Inspection & Enforcement…



And, What If it was more than just firewalling…

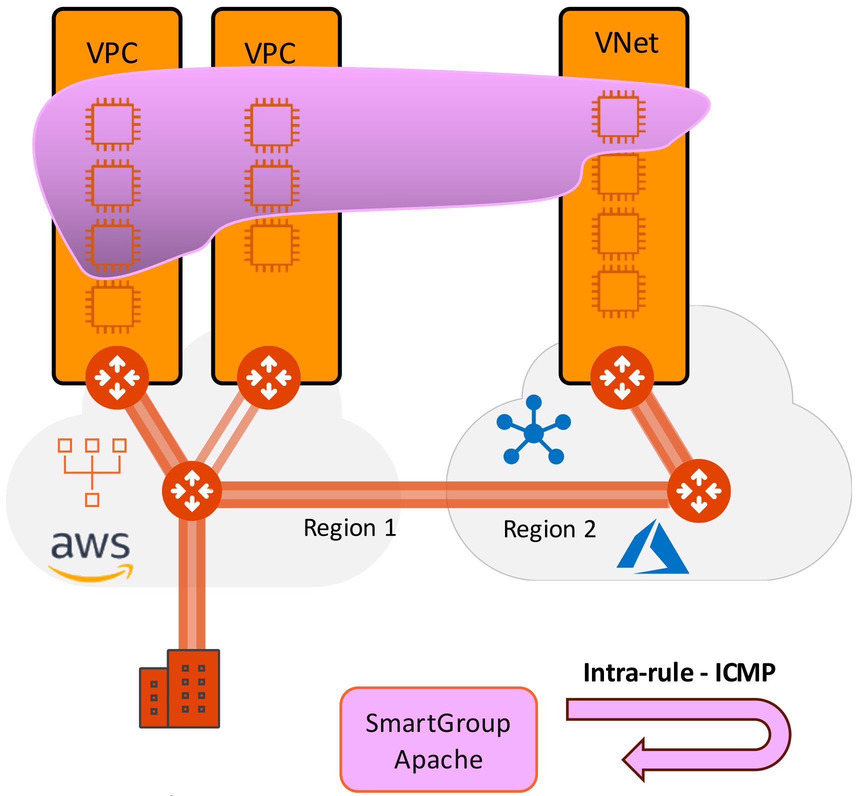


Policy Creation Looked Like One Big Firewall … A Distributed Cloud Firewall…



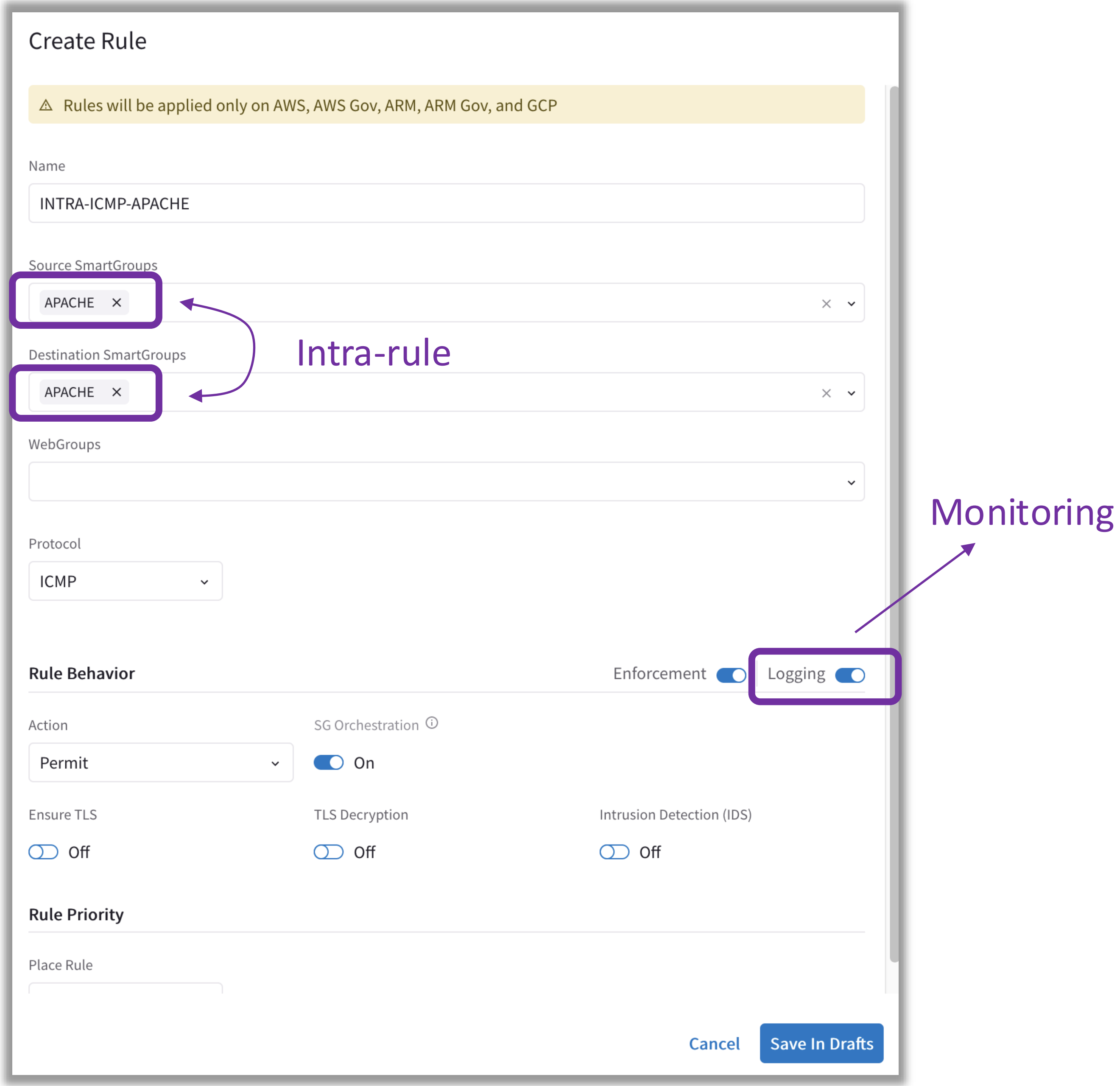
Where and How Policies Are Enforced Is Abstracted…

Micro-Segmention: SmartGroups, Intra-Rules and Inter-Rules (1)

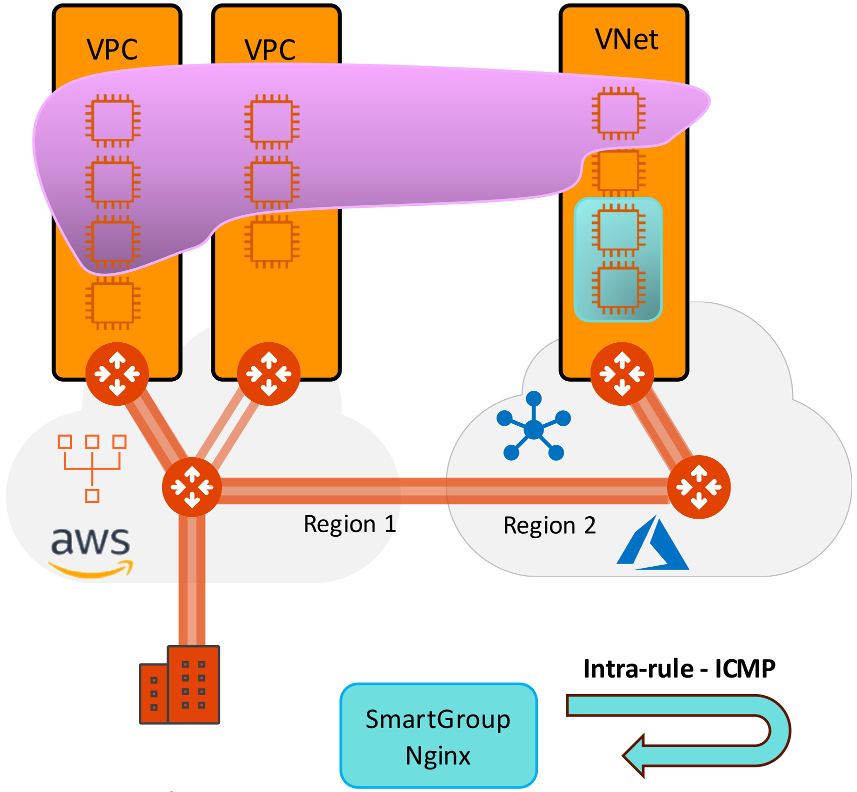


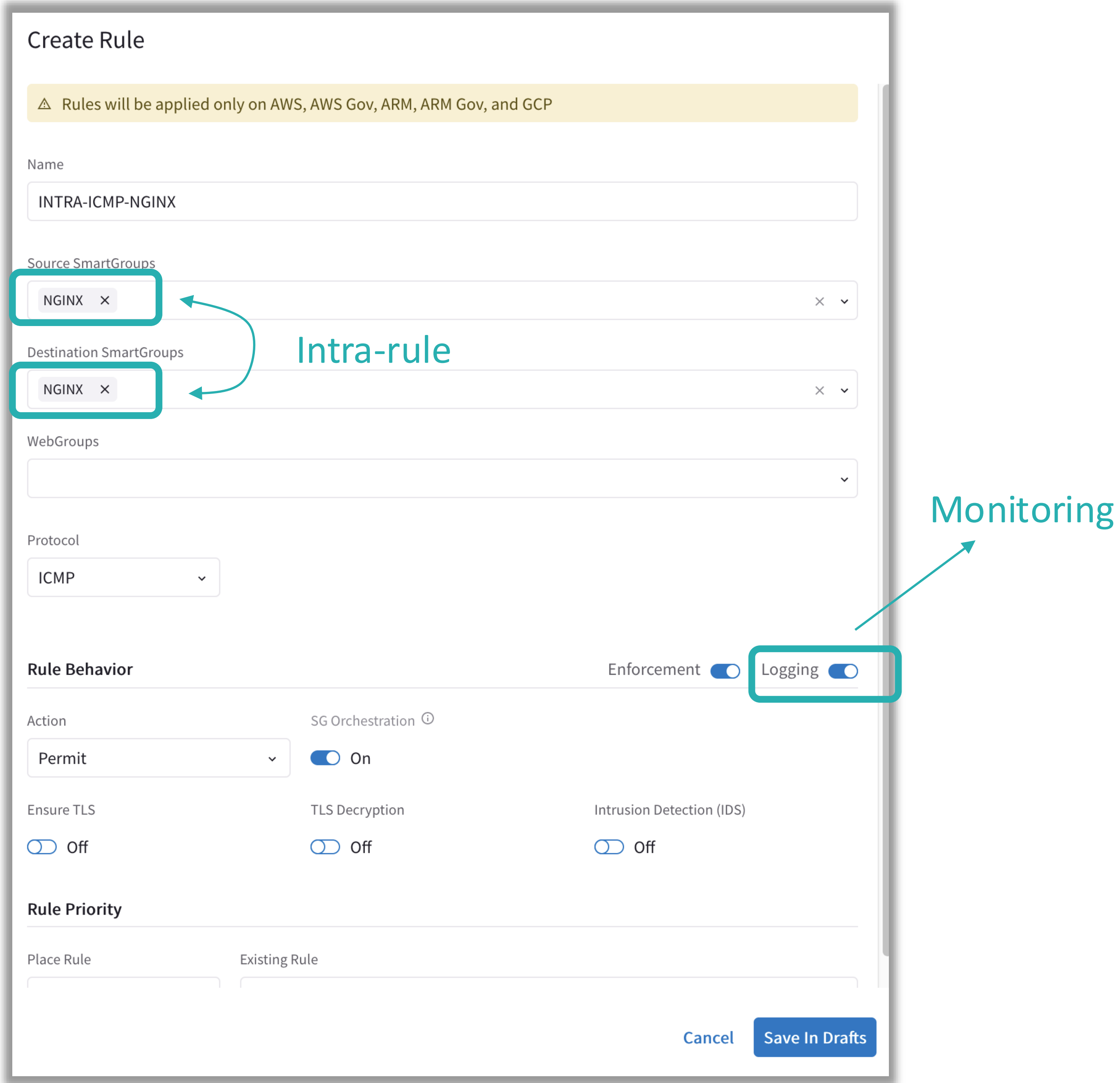
Scenario #1

- **Create a DCF rule for the APACHE SmartGroup with the following requirements:**
  - Permit ICMP traffic internally
  - Enable the Logging feature



Micro-Segmention: SmartGroups, Intra-Rules and Inter-Rules (2)

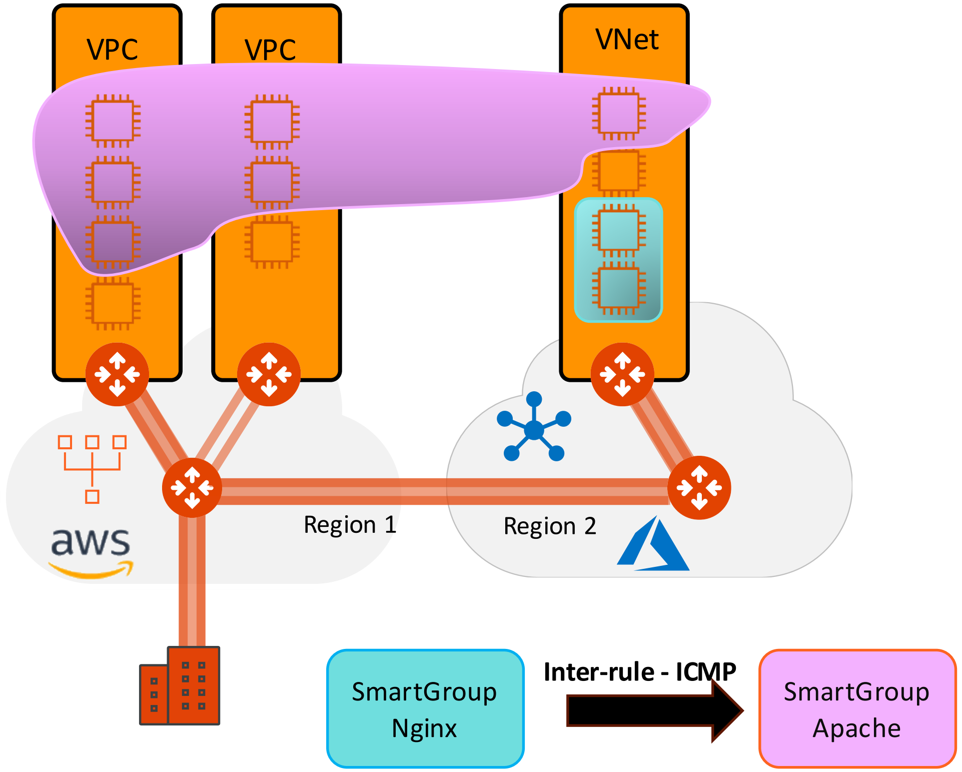


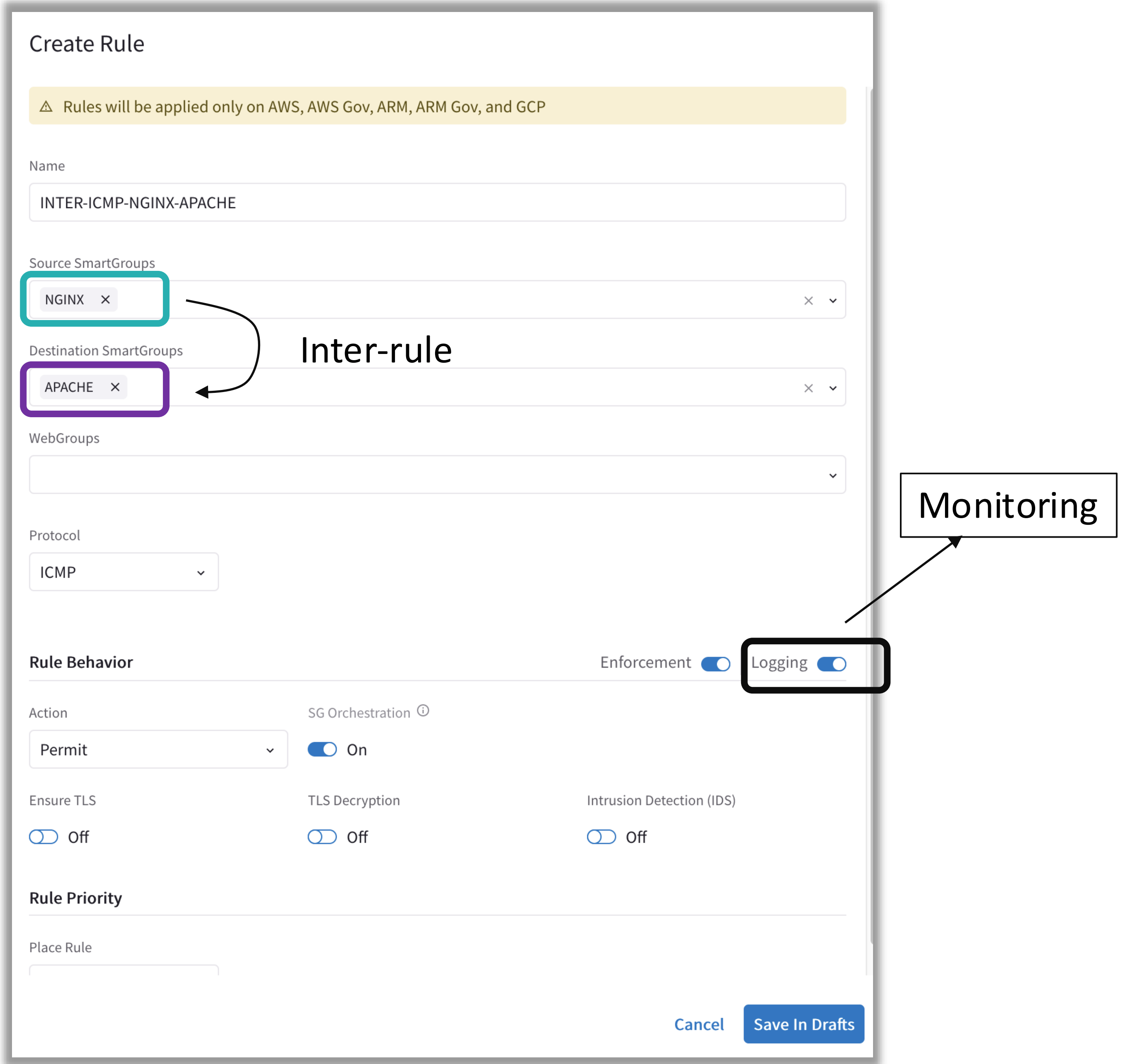


Scenario #2

- **Create a DCF rule for the NGINX SmartGroup with the following requirements:**
  - Permit ICMP traffic internally
  - Enable the Logging feature

Micro-Segmention: SmartGroups, Intra-Rules and Inter-Rules (3)

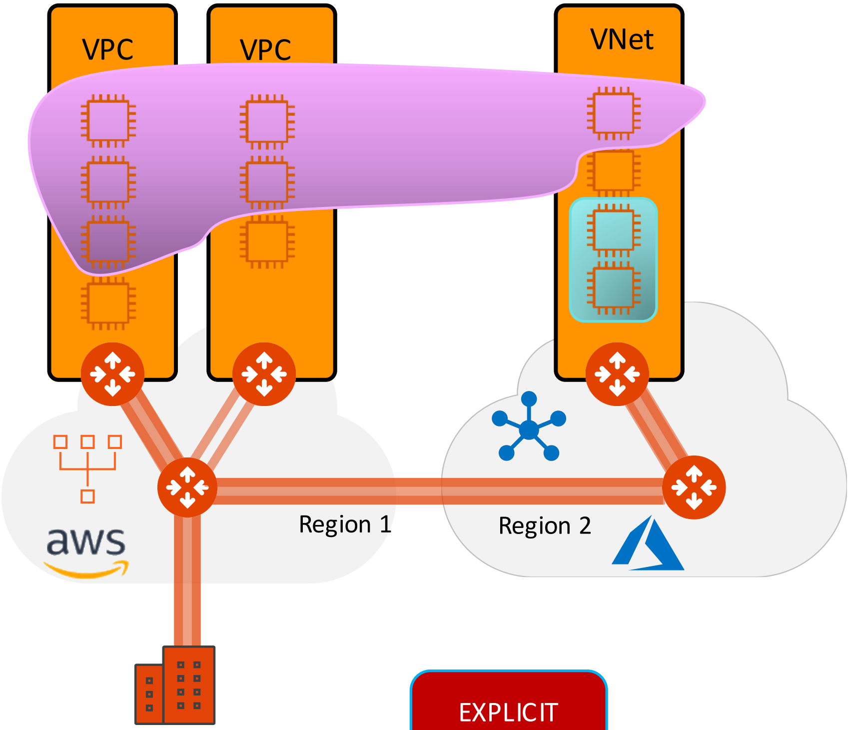




Scenario #3

- **Create a DCF rule from the NGINX SmartGroup towards the APACHE SmartGroup, solely, not the inverse (NO bidirectional!), with the following requirements:**
  - Allow ICMP traffic between the two SGs
  - Enable the Logging feature

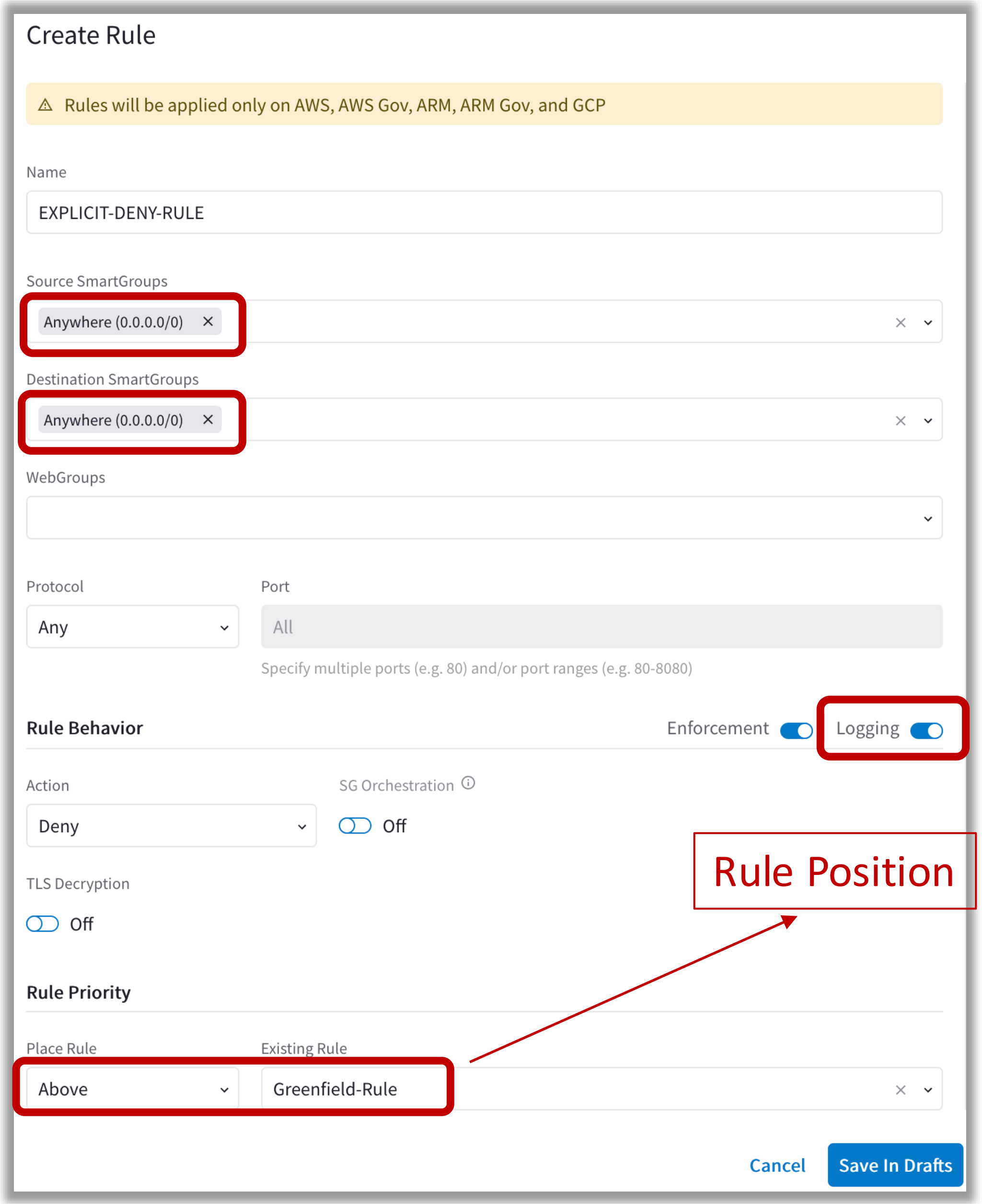
Micro-Segmention: SmartGroups, Intra-Rules and Inter-Rules (4)



EXPLICIT DENY

Scenario #4

- **Create a DCF rule that explicitly deny any kind of traffic based on the following requirements:**
  - Insert the rule below the previous created rules and above the Greenfield-Rule
  - Enable the Logging feature



Monitoring

Rule Position

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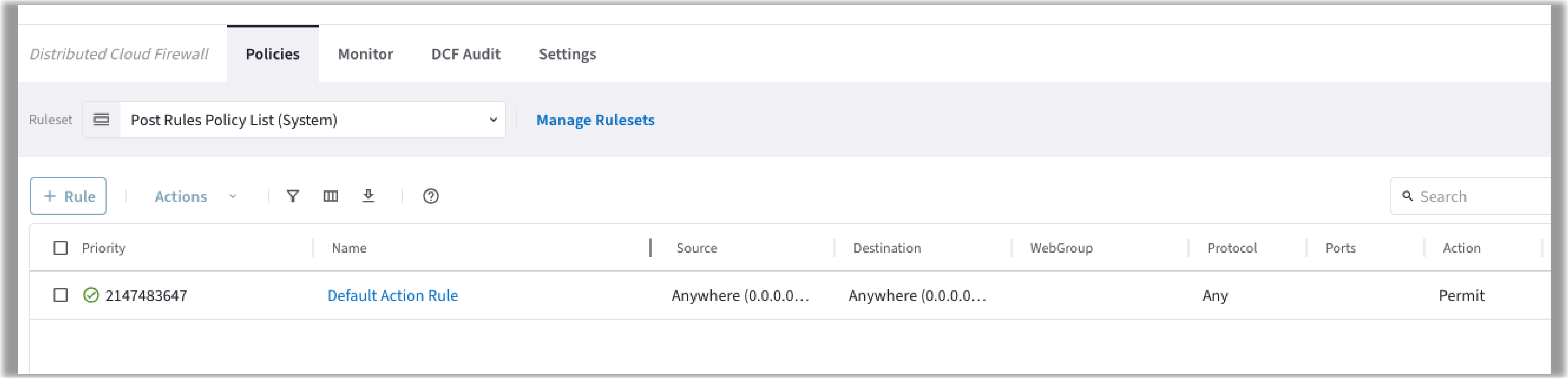
Tools for troubleshooting Distributed Cloud Firewall

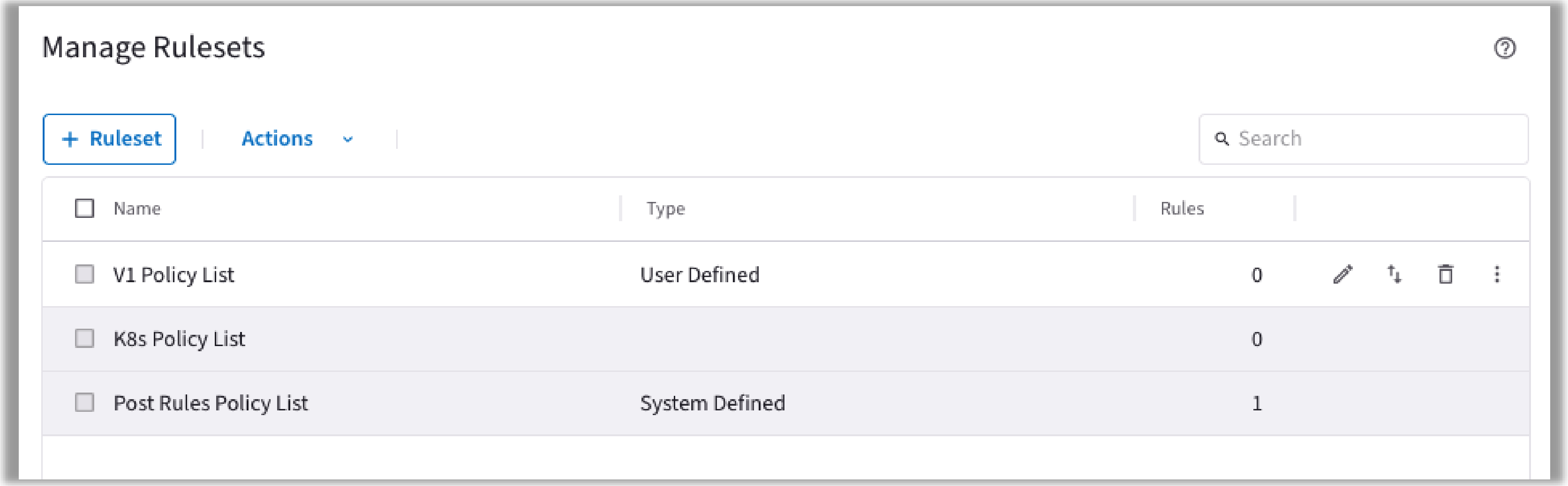
Rulesets – Post Rules Policy List (System)

- When you enable the Distributed Cloud Firewall (DCF), the system automatically installs a **Default Action Rule** (Any-Any-Any: Permit) within the **Post Rules Policy List**.
- **Understanding the Default Action Rule:** This is a mandatory system rule that cannot be deleted. It functions as the final policy in the evaluation chain, governing any traffic not explicitly addressed by your user-defined rules.
- **Enforce Zero Trust:** To align with Zero Trust principles, you must edit this rule immediately after creation. Change the action from **Permit** to **Deny** to ensure that traffic is blocked by default if it does not match a specific allow rule.



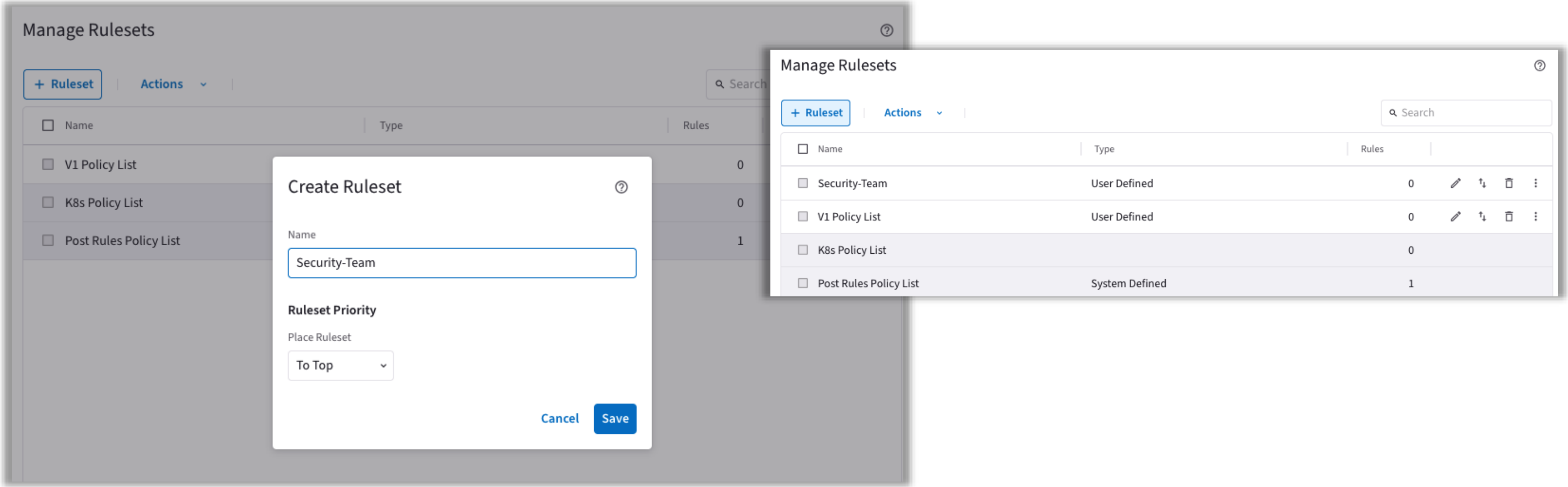
Rulesets – V1 Policy List

- The **V1 Policy List** is the centralized workspace where you define and manage your custom security policies. Because this list dictates the order of operations for your firewall, effective collaboration is essential.
- **Foster Cross-Team Collaboration:** Engage key stakeholders across your organization to align on security requirements and operational needs.
- **Establish Rule Precedence:** Collectively determine the hierarchy of your rulesets. This ensures that the most critical, high-priority security policies are evaluated first, maintaining a robust and predictable security posture.



Creating a Distributed Cloud Firewall Ruleset

- **How to Create a Ruleset**
- *Follow these steps to structure your security policy hierarchy:*
- Go to **Security > Distributed Cloud Firewall > Policies** and click **Manage Rulesets**.
- **Initialize Creation:** In the *Manage Rulesets* dialog, click the **+ Ruleset** button to open the configuration window.
- **Configure Your Ruleset:** Provide the necessary details for your new ruleset



Rulesets – Post Rules Policy List (System)

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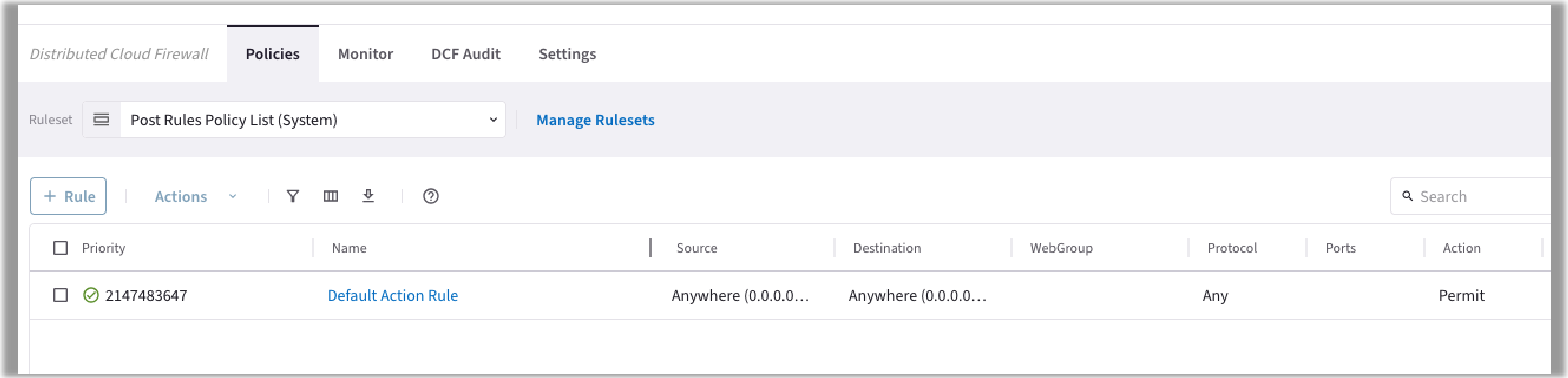
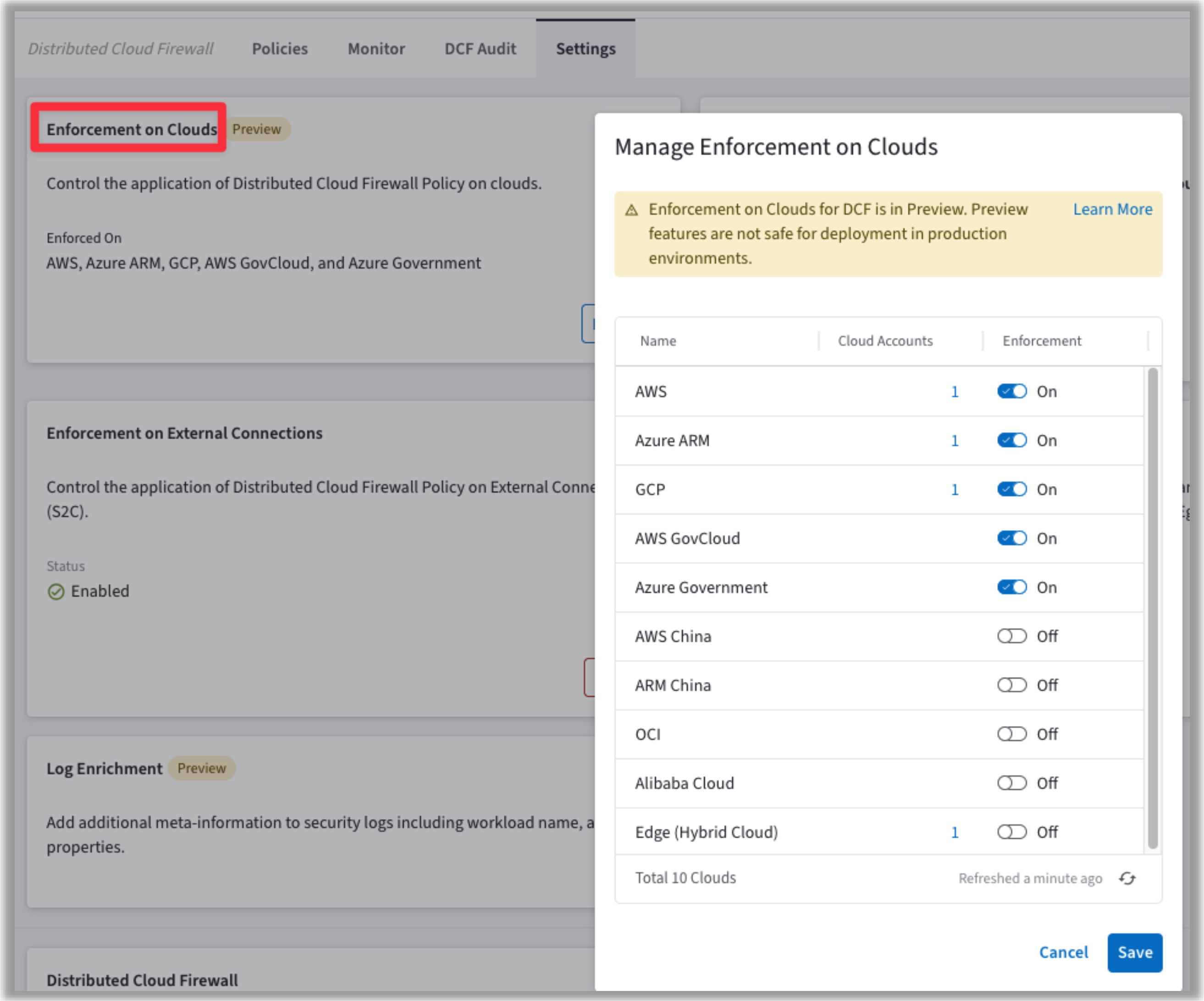


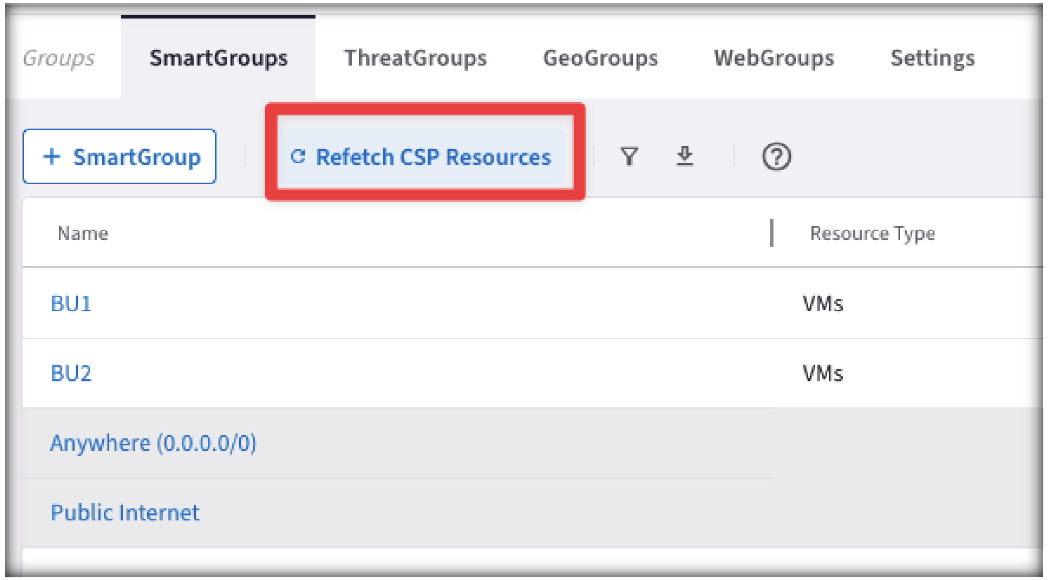
Enforcement on Clouds

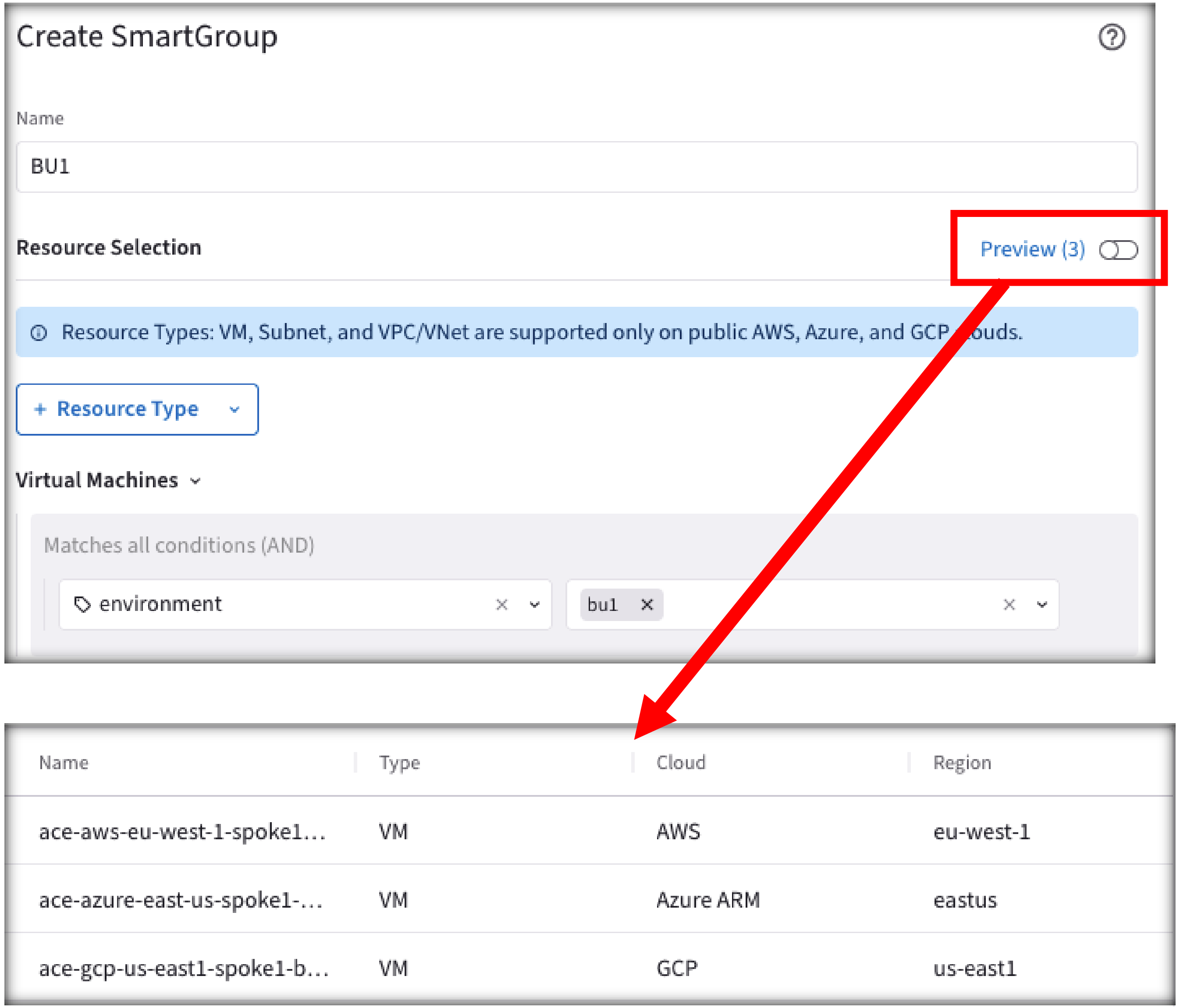
- **Advanced Enforcement Control**
- Although enforcement is applied by default to the relevant gateways within the CNSF, you have the flexibility to further refine where your policies are enacted. Through the **Manage Cloud Execution** settings, you can define specific constraints to dictate exactly which Cloud Service Provider (CSP) environments should enforce your policies.



Creation of the SmartGroup: the right matching criteria dilemma

- Choose the right matching criteria for resources that you want to see assigned to a specific SmartGroup:
- Use the **Preview Resources** toggle switch to verify the selected resources that have been mapped to the Smart Group
- Use the On-Demand **Refetch CSP Resources** button to retrieve the most recent inventory

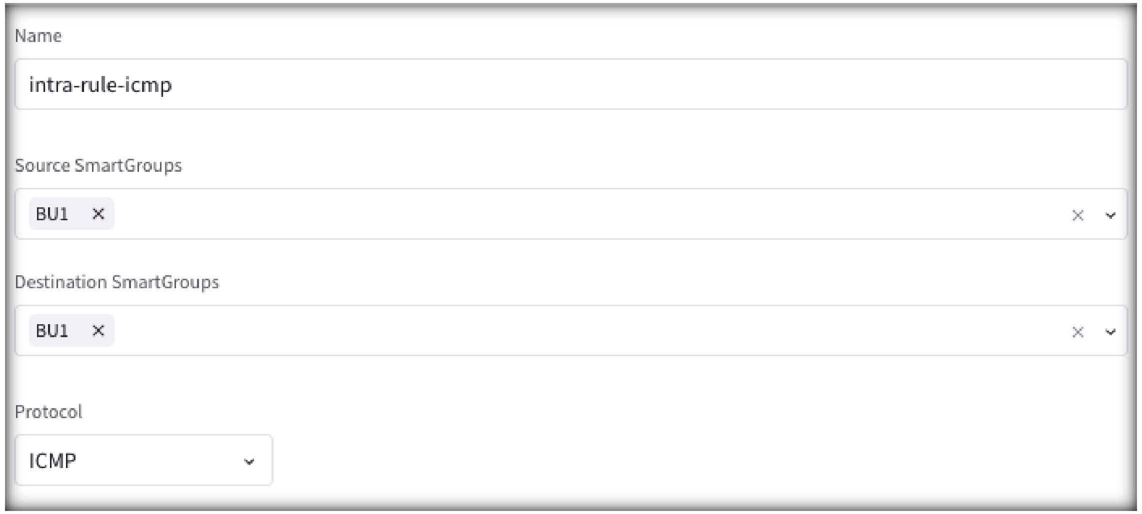




Creation of the DCF Rules: intra-rule vs. inter-rule

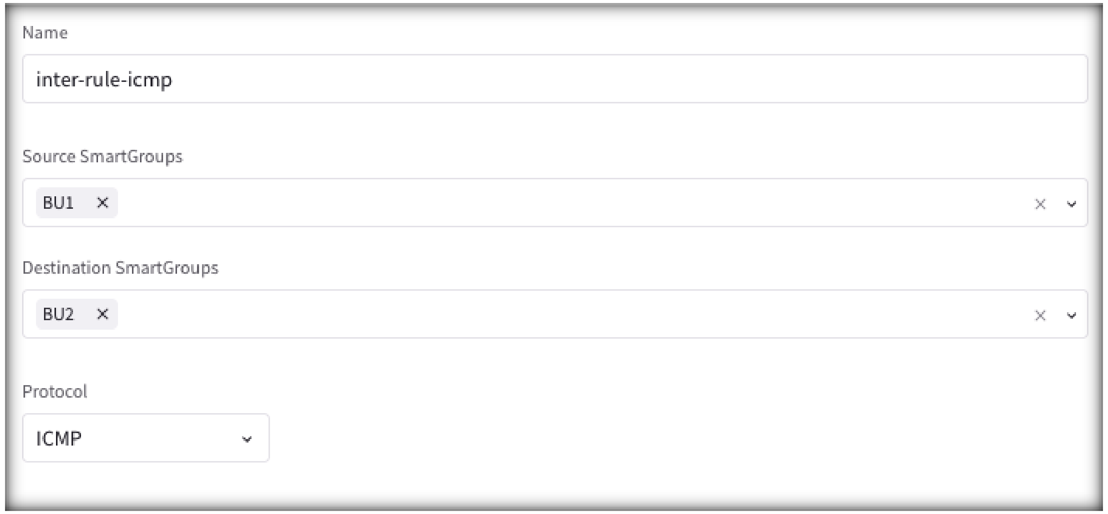
1) **Intra-rule** will affect the traffic WITHIN a Smart Group

- ❑ Source Smart Group and Destination Smart Group must be the same



2) **Inter-rule** will affect the traffic BETWEEN Smart Groups

- ❑ Source Smart Group and Destination Smart Group must differ



Next:

Lab 8 Distributed Cloud Firewall